

Topic: Basic Organization of Computer -- Self-Learning

Q. Discuss the basic organization of a computer system. Explain the primary functions of the Input, Output, Memory, ALU, and Control Units.

[DEF] Definition to Remember

The Basic Organization of a Computer relies on five fundamental, interconnected units that facilitate the flow of data: Input Unit, Output Unit, Memory Unit, Arithmetic and Logic Unit (ALU), and Control Unit (CU). Together, they accept, store, process, and output data.

Basic Organization of Computer (Self-Learning)

Unit	Primary Function	Examples
1. Input Unit	Accepts human-readable data/instructio	Keyboard, Mouse, Scanner
2. Output Unit	Retrieves binary data from Memory, tra	Monitor, Printer, Speaker
3. Memory Unit	Stores data and instructions. Divided	RAM, SSD
4. Arithmetic & Logic Unit (ALU)	The execution engine. Performs math (a	Inside CPU
5. Control Unit (CU)	The central nervous system. Coordinate	Inside CPU

2. The Central Processing Unit (CPU)

The CPU is the "brain" of the computer. It is physically made up of:

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3. Basic Data Flow

1. User types data -> Input Unit -> translates to binary.

2. Da

[MUST-WRITE] Must-Write Points (for 10 marks)

1. A computer is organized into 5 units: Input, Output, Memory, ALU, Control Unit.
2. Input Unit: Converts external data to binary and feeds it to the system.
3. Output Unit: Converts binary processed data back to human-readable form.
4. Memory Unit: Stores data/programs (Primary = RAM; Secondary = Hard Drive).
5. ALU (Arithmetic Logic Unit): Performs actual math and logical processing.
6. CU (Control Unit): Manages/coordinates all units (Fetches, Decodes, executes).
7. The CPU is the combination of the ALU, CU, and Registers.

[RECALL] Quick Recall

Computer Org -> Input (data in) -> Memory (storage) -> CU (manager) -> ALU (processor) -> Output (data out) -> CPU = CU + ALU + Registers